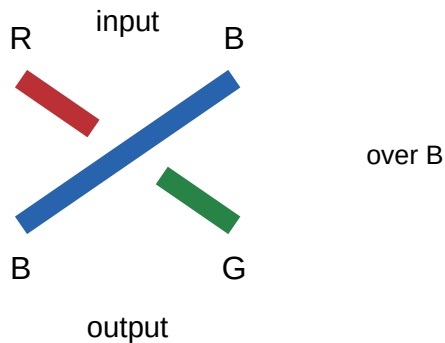
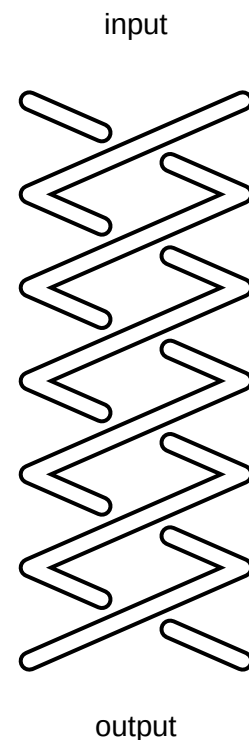


Use red (R), blue (B), and green (G). Give each whole arc one color; an arc continues through overpasses and stops at underpasses or open ends. At every crossing, the three colors are all the same or all different. Read machines from top inputs to bottom outputs.

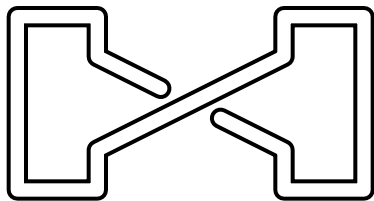


Problem 1: Feed every ordered pair of R, B and G into repeated copies of this crossing. How many crossings first return the same left-right pair? Can any pair fail to return? Explain your rule.

	first return after ____ crossings
R R	_____
R B	_____
R G	_____
B R	_____
B B	_____
B G	_____
G R	_____
G B	_____
G G	_____

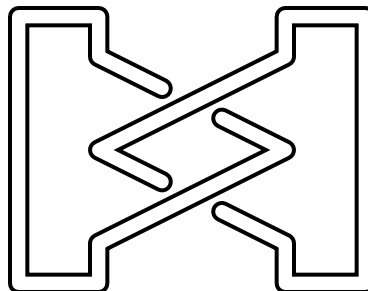


Problem 2: Each picture joins left output to left input, and right output to right input, outside the crossing strip. Count every valid coloring, including one-color choices. Which pictures have equal counts but different numbers of separate cords? Explain why your counts are complete.



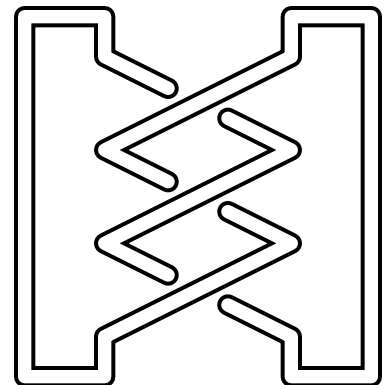
1 crossing

colorings: ____ cords: ____



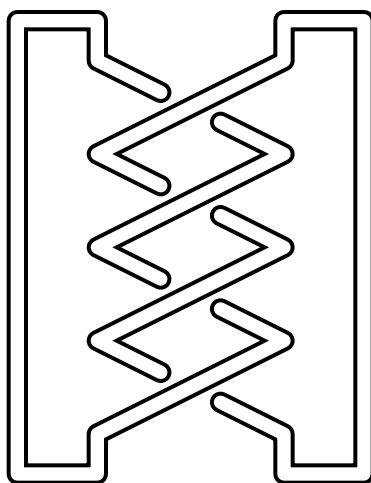
2 crossings

colorings: ____ cords: ____



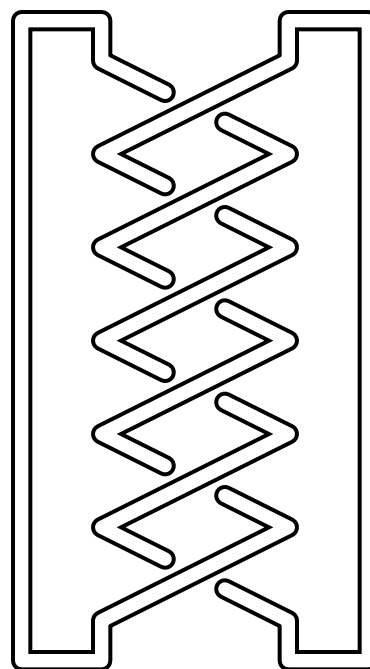
3 crossings

colorings: ____ cords: ____



4 crossings

colorings: ____ cords: ____



6 crossings

colorings: ____ cords: ____

To join two closed diagrams, cut the marked crossing-free pieces and join upper ends to upper ends, lower ends to lower ends. Joined ends must have the same color.

Problem 3: How many valid colorings does each joined picture have? Predict the count if three three-crossing loops are joined. Explain your counts.

